CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

8.2 Diffraction

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

Which of the following is the definition of diffraction?

A change of direction when waves cross the boundary between one medium and another

B splitting of white light into colours

C addition of two coherent waves to produce a stationary wave pattern

D bending of waves around an obstacle

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges

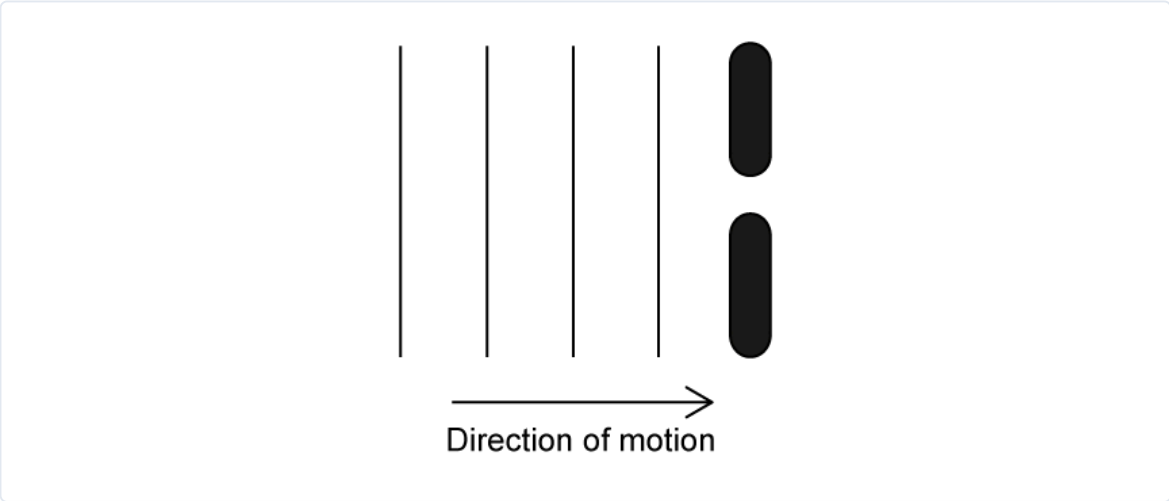
A is incorrect because this definition describes refraction

B is incorrect because this definition describes dispersion

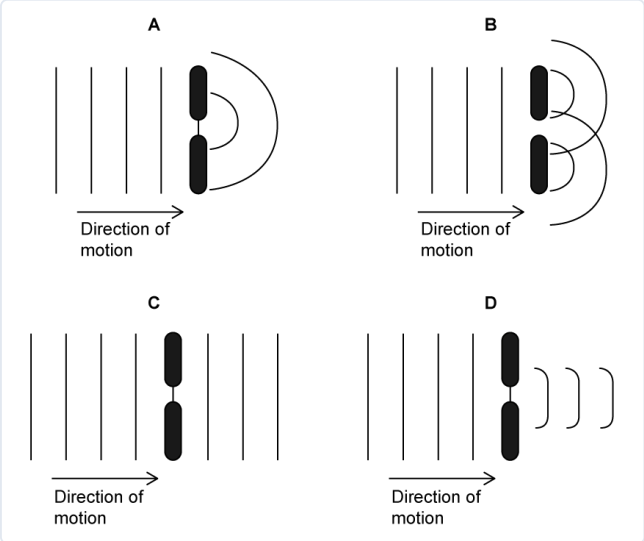
C is incorrect because this definition describes superposition

Question 2 Easy [1 mark]

The diagram shows a wave approaching an aperture.



Which diagram correctly shows the resulting diffraction pattern for the wave?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The wave would diffract through the aperture
    - This discounts options **B** and **C**

- The wavelength is larger than the size of the aperture
    - Therefore the diffraction would be great
    - This discounts option **D**
- Option **A** is the only answer that fits these criteria

B is incorrect as this pattern is the double-slit diffraction pattern that cannot be formed from a single aperture

C is incorrect as no diffraction is occurring in the diagram. This would only happen if the wavelength was very much smaller than the aperture

D is incorrect as the wave is not diffracting very much. This is the pattern we would expect to see if the wavelength was smaller than the aperture, but the wavelength is actually larger than the aperture

Question 3 Easy [1 mark]

For which types of wave does diffraction occur?

A In all waves

B In soundwaves only

C In transverse waves only

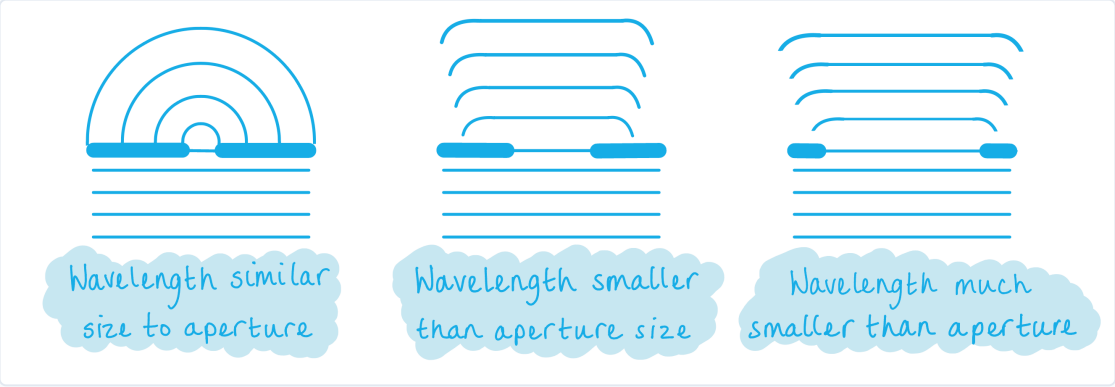
D In light waves only

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Diffraction is the spreading out of a wave around an obstacle or aperture
- Diffraction occurs in **all types of wave**
    - Including sound waves, light waves and all transverse waves

Diffraction is greatest when the wavelength of the wave is of a **similar size** to the aperture, or **larger** than the aperture.

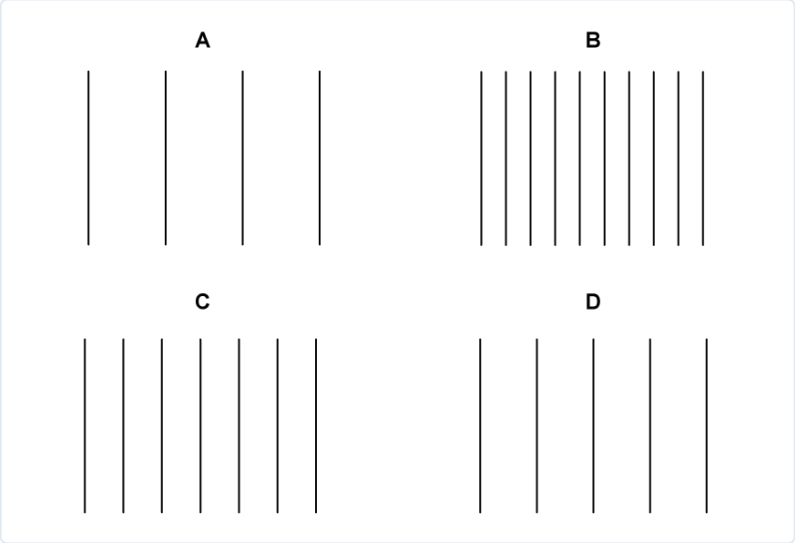


Sound waves are easily diffracted around open doors (the human voice has a wavelength very similar in size to the width of an open door). Long wave radio signals (with wavelengths of around a kilometre) are diffracted around most objects so that the signal can be received in most places. However, shorter wave radio signals (with wavelengths of around one metre) do not diffract significantly around hills and cliffs leaving areas of 'shadow' where the signals cannot be received. To observe diffraction in visible light (with wavelengths of 400 - 700 nm) the apertures used need to be similar in size to the wavelength.

Question 4 Easy [1 mark]

Wavefront diagrams can be used to indicate the frequency of waves.

Which wave has the lowest frequency?



A

B

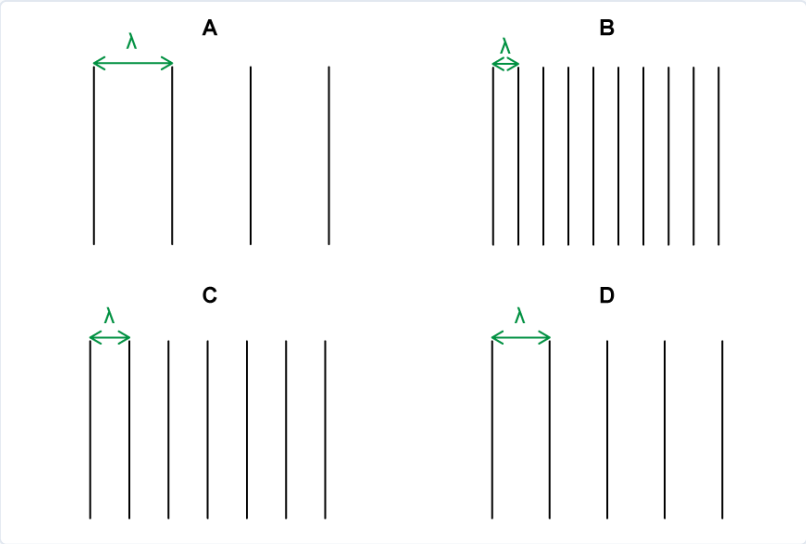
C

D

Mark Scheme & Explanation:

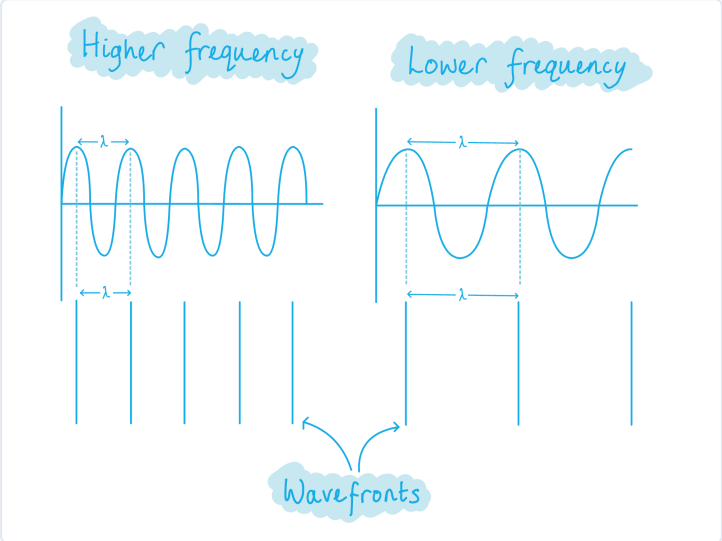
*The correct answer is **A** because:*

- Wavefronts are lines joining all the points on a wave that oscillate in-phase, such as crests or troughs
  - The distance between two wavefronts is equal to one wavelength
- Lower frequency waves have longer wavelengths, and higher frequency waves have shorter wavelengths
- Wave **A** has the longest wavelength
  - Therefore, wave **A** has the lowest frequency



Wavefront diagrams are a model, a simplified version of a scenario that allows us to focus on the most relevant aspects of a situation. It is helpful to think of wavefront diagrams as a snapshot in time over a specific area that shows you where the wavefronts are at that moment. Higher frequency waves will have more wavefronts in their snapshot than lower frequency waves.

In this way, we can use wavelength as a guide to determine the frequency from a wavefront diagram.



Question 5 Easy [1 mark]

Which answer option gives a correct description of wave diffraction?

A A wave is returned from its original direction when incident on a shiny surface

B A wave changes direction when it passes between materials of different density

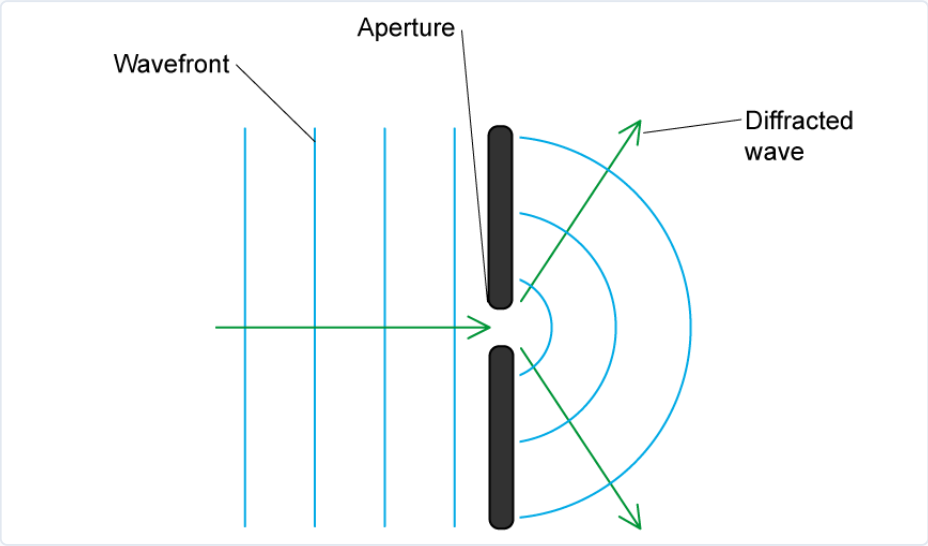
C Waves spreading out when they pass an obstruction

D The constructive interference of two waves in phase

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Diffraction is the **spreading out** of waves, as shown in the diagram below



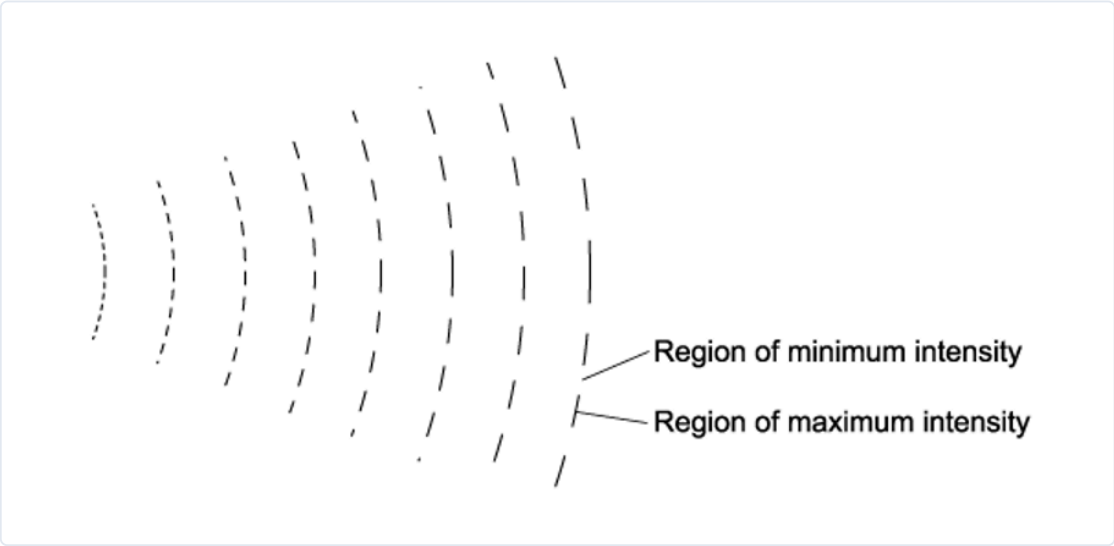
A is incorrect as this is describing reflection

B is incorrect as this is describing refraction

D is incorrect as this is describing a situation of interference

Question 6 Medium [1 mark]

The diagram shows the pattern of waves. The source of the waves was unable to be seen.



Which of the following would cause this pattern?

A diffraction only

B interference only

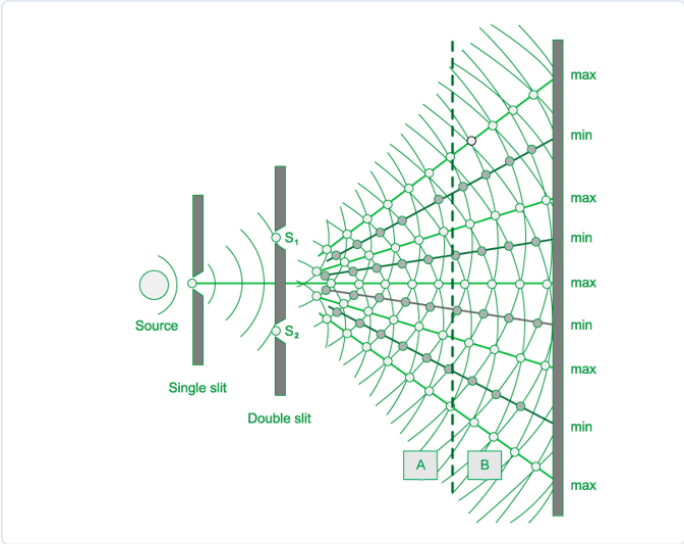
C coherence only

D diffraction and interference

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- When the light is passed through a double slit two coherent sources of waves are produced that interfere
- Light diffracts from each slit, because the slits are narrow
- These waves overlap and interfere constructively (maximum intensity) and destructively (minimum intensity)



A is incorrect because diffraction through a single slit and would not produce the pattern seen

B is incorrect because the light would need to be diffracted by a single slit to produce the wave pattern seen

C is incorrect because coherence is when the frequency and waveform are identical, and the phase difference is constant. A coherent wave would not produce this wave pattern

Question 7 Medium [1 mark]

An atomic lattice has a spacing of around 10^{-10} m, which electromagnetic wave would cause the most significant diffraction effect?

A X-ray

B infra-red

C microwave

D ultraviolet

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges
- For the largest diffraction effect to be seen the wavelength needs to be close to the aperture size
    - X-rays have a wavelength of 10^{-8} to 10^{-13} m

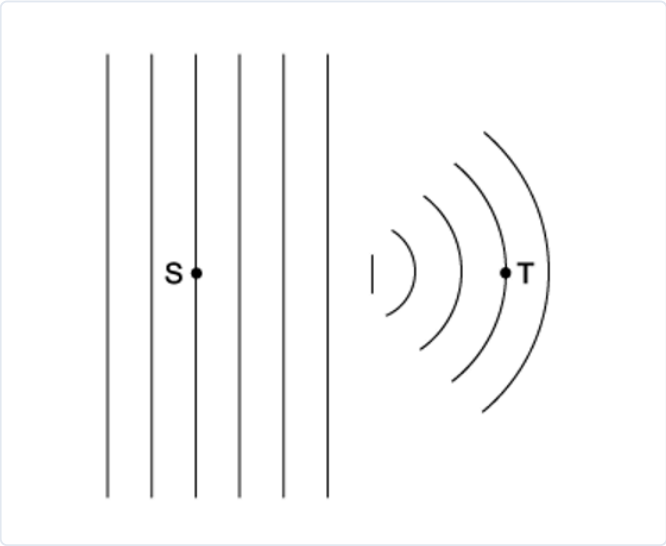
B is incorrect because infra-red waves have a wavelength of 10^{-3} to 10^{-7} m

C is incorrect because microwaves have a wavelength of 10^{-1} to 10^{-3} m

D is incorrect because ultraviolet waves have a wavelength of 10^{-7} to 10^{-8} m

Question 8 Medium [1 mark]

A ripple tank was used to demonstrate plain wavefronts passed through a gap, as shown in the diagram.



Which property would be different at S compared to T?

A frequency

B amplitude

C wavelength

D velocity

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Frequency and wavelength are related to each by the equation : $v = f\lambda$
- The diagram shows the wavefronts at S and T, the only difference between the two points is that the wave has travelled through a slit or gap
- Only the amplitude of the wave changes as some energy is dissipated as it passes through the gap

A is incorrect because the frequency is determined by the source and so remains constant

C is incorrect because the wavelength is constant as the wavefronts are constant in the diagram

D is incorrect because the velocity would remain constant as both wavelength and frequency are constant

Question 9 Medium [1 mark]

When a wave passes an obstruction diffraction can be observed, this effect is greatest when the wavelength and obstruction are similar in size.

When waves are travelling through the air, which of the following would best demonstrate diffraction?

A radio waves passing a copper wire

B visible light waves passing a gate post

C microwaves passing a steel post

D sound waves passing human hair

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges
- For the largest diffraction effect to be seen the wavelength needs to be close to the aperture size
    - Microwaves have a wavelength of 1 m to 10^{-3} m so would be a similar size to the steel post

A is incorrect because radio waves have a wavelength of 10^{-1} to 10^{6} m so would not be diffracted by the copper wire

B is incorrect because visible light has a wavelength of $7 - 4 \times 10^{-7}$ m so would not be diffracted by the gate post

D is incorrect because sound waves have a wavelength of 1.7×10^{-2} m to 17 m so would not be diffracted by the human hair

Question 10 Medium [1 mark]

A slit of width 0.2 m has sound waves incident upon it. The speed of sound is 340 ms^{-1}.

What does the frequency of the sound wave need to be for the greatest diffraction to occur?

A 500 Hz

B 1000 Hz

C 1700 Hz

D 3000 Hz

Mark Scheme & Explanation:

*The correct answer is **C** because*

- The greatest diffraction occurs when the gap is the same size as the wavelength going through it
- The wave equation is
  - $v = f\lambda$
  - $\lambda = \frac{v}{f}$
- When the frequency is 1700 Hz
  - $\lambda = \frac{340}{1700} = 0.2\ \text{m}$
- This is option **C**

Question 11 Hard [1 mark]

Where wavelength is given as λ, and w is the width of the aperture, which of the following answer options would result in the greatest diffraction?

A $\frac{\lambda}{w} \approx 1$

B $\frac{\lambda}{w} < 1$

C $\frac{\lambda}{w} \ll 1$

D $\frac{w}{\lambda} > 1$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The greatest amount of diffraction occurs when the wavelength is equal to the size of the aperture

B is incorrect because when the size of the aperture is larger than the wavelength, less diffraction will occur

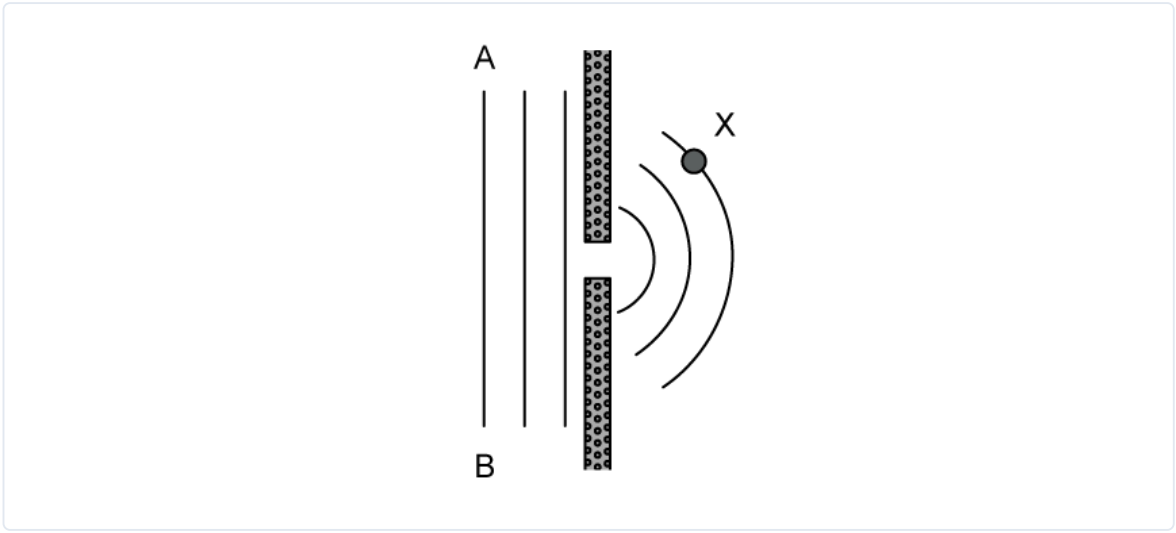
C is incorrect because when the size of the aperture is much larger than the wavelength, much less diffraction will occur

D is incorrect because when the wavelength is smaller than the size of the aperture, less diffraction will occur



Question 12 Hard [1 mark]

The diagram shows wavefronts in a ripple tank diffracting through an aperture. The waves have a speed, *c*, and a wavelength, λ.



Which of the following expressions gives the time taken for the wavefront AB to reach point X?

A $\frac{3\lambda}{c}$

B $\frac{6\lambda}{c}$

C $\frac{3\lambda}{3c}$

D $\frac{5\lambda}{c}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- $v = \frac{s}{t} \rightarrow t = \frac{s}{v}$
- Distance, *s*, is equal to the number of wavelengths between AB and X
  - Therefore, $s = 5\lambda$
- Speed, v, is equal to the speed of the wave
  - Therefore, $v = c$
- Hence, the time taken is $t = \frac{5\lambda}{c}$

- This is answer option **D**

Question 13 Hard [1 mark]

A 5 cm cube casts a shadow when a torch is shone from behind it. When a speaker is placed behind it emitting sound of a frequency 4000 Hz, the sound can still be detected behind the cube.

Which of the following statements explain this observation?

A Light travels faster than sound

B Light is a transverse wave and sound is longitudinal

C Light waves have shorter wavelengths than sound waves

D Light waves are electromagnetic and sound waves are compression waves

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The sound waves have much longer wavelengths than the width of the cube
- For sound, $v = f\lambda \rightarrow \lambda = \frac{v}{f} = \frac{340}{4000} = 0.085 \text{ m}$
  - 8.5 cm is close to 5 cm significant diffraction will occur
- For light, the wavelength of the visible spectrum at 700 nm to 400 nm
  - $700 \text{ nm} \ll 5 \text{ cm}$ therefore no diffraction will occur

Question 14 Hard [1 mark]

Sound diffracts around a loudspeaker aperture of diameter 15 cm.

The speed of sound is 340 m s^{-1}, and the frequency of the sound is 3 kHz.

What effect would be observed?

A No diffraction

B Slight diffraction

C Significant diffraction

D Not enough information to tell

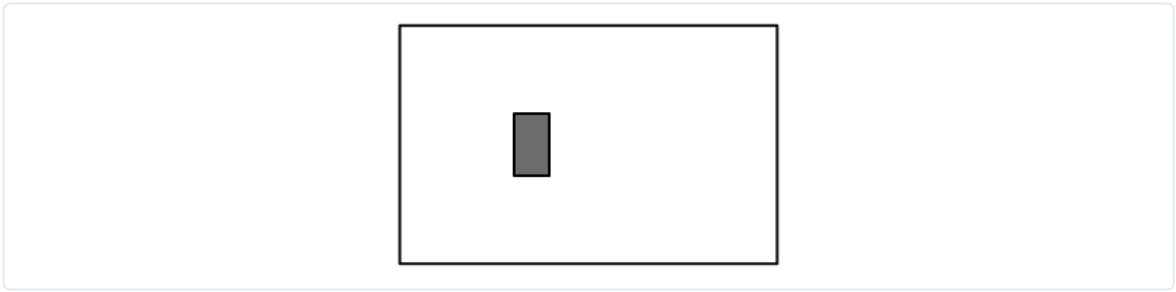
Mark Scheme & Explanation:

*The correct answer is **C** because:*

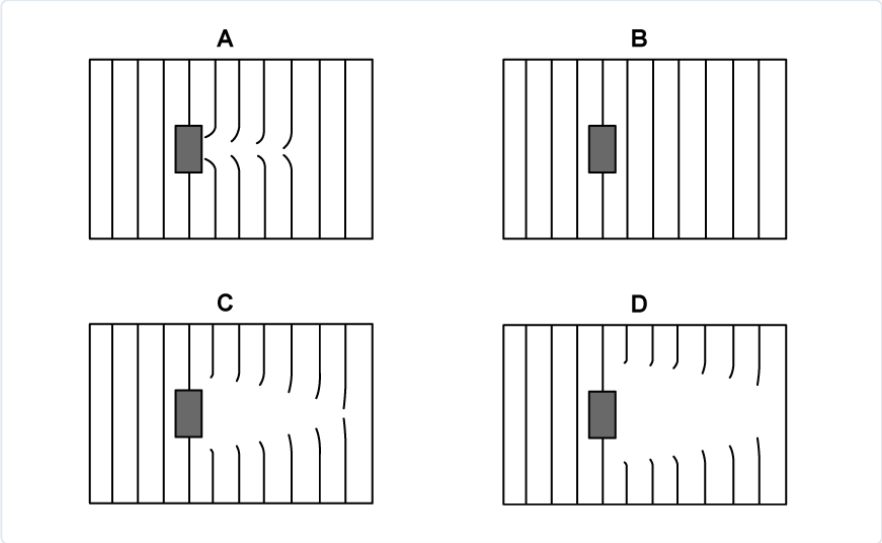
- Significant diffraction occurs when the size of the aperture is similar to the wavelength of the wave
- $v = f\lambda \rightarrow \lambda = \frac{v}{f} = \frac{340}{3000} = 0.11 \text{ m}$
- $11 \text{ cm} \approx 15 \text{ cm}$ therefore significant diffraction would occur

Question 15 Hard [1 mark]

A block is placed in a ripple tank as shown in the diagram.



Which answer option shows the effect on the wavefront pattern observed?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- When a wavefront meets the obstacle, it diffracts around the edges
- Behind the obstacle a 'shadow' forms where the wave is blocked
- When the wavelength of the wave is similar to the width of the object, significant diffraction occurs
    - This means that the waves join back up after the 'shadow'
- When the wavelength is significantly smaller than the width of the object, the diffraction is less, so the shadow is longer

- Therefore, **A** is the only correct answer option

